N3 — Spectral Concentration of the Apex Unitary: A Structural Reduction

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Abstract

For each integer $N \geq 2$ let $U_N \in U(N-1)$ be the apex unitary intertwining the Helmert and character frames of the mean-zero hyperplane of $\mathbb{C}^N$. Numerically, the eigenvalues of U_N concentrate at the four eigenvalues $\{1, i, -1, -i\}$ of the discrete Fourier transform, with multiplicities that depend only on N modulo 4. We prove that this concentration is governed by a single non-degeneracy. Writing $U_N^{-1} = U_N^*$, the eigenvalues of U_N all lie strictly within an eighth of a turn of a fourth root of unity if and only if

$$\det(U_N^4 + I) \neq 0, \qquad \text{equivalently} \qquad \left\|U_N^2 - U_N^{-2}\right\|_{\mathrm{op}} < 2,$$

and we give the exact identity $\|U_N^2 - U_N^{-2}\| = 2\sin(4\pi\delta_{\max})$, where $\delta_{\max}$ is the largest displacement of an eigenvalue from its nearest fourth root. Around this reduction we establish, unconditionally, the supporting structure: the determinant $\det(U_N^4 + I) = (-1)^{\binom{N-1}{2}} \det(U_N^2 + U_N^{-2}) = 2^{N-1}\prod_j \cos(4\pi\delta_j)$ is real; the cluster multiplicities are recovered exactly by rounding the first four power traces; the field trace of $\operatorname{tr} U_N$ over $\mathbb{Q}(\zeta_N)$ is computed in closed form; and a rescaling of the matrix entries produces an explicit half-line integral operator $\mathcal{U} = (A - I)\Phi$ (A the Hardy mean, Φ the half-line Fourier transform) whose leading eigenvalue is the displacement of the single edge mode that controls everything. The spectral-concentration theorem thus reduces to one isolated estimate, $\delta_{\max} < \frac{1}{8}$, equivalently $\det(U_N^4 + I) \neq 0$ for all N, which the numerics support with margin (the limiting displacement is $\approx 0.097 < 0.125$).

Contents

1. Introduction 2
2. The apex unitary and the Fourier spectrum 3
3. The concentration phenomenon 3
4. The no-crossing reduction 4
5. Power-trace formulations 5
6. The edge operator 6
7. The trace and its field trace 6
8. The structural reduction 7

References 8

1. Introduction

The companion papers study the apex unitary U_N as an arithmetic object: [N1] sets up its definition over the multiquadratic–cyclotomic compositum and its Galois action, and [N2] proves the closed form $\det U_N = i^{-\binom{N-1}{2}}$ via the factorization $\det U_N = \det H \cdot \det F$ into a Helmert sign and Schur's Fourier determinant. The present paper turns to the full spectrum.

The starting observation is that U_N is, by construction, a *Helmert deformation of the discrete Fourier transform*. It is the nontrivial diagonal block of $M = HF^{\mathsf{T}}$, where H is the full normalized Helmert matrix and F the symmetric Fourier matrix (N2, Prop. 3.1). The Fourier matrix is the metaplectic object par excellence: $F^4 = I$, so its spectrum is contained in $\{1, i, -1, -i\}$, with the four multiplicities classical (Schur; McClellan–Parks; the eigenvectors approximate Hermite functions). Numerically, the eigenvalues of U_N inherit this fourfold structure — they cluster tightly at the same four points — even though H is a large rotation, not a small perturbation. This paper makes that statement precise and reduces it to a single, sharply-stated estimate.

The concentration. Write the eigenvalues of U_N as $\lambda_j = e^{2\pi i\theta_j}$, let i^{c_j} be the nearest fourth root of unity, and let $\delta_j = \theta_j - \frac{c_j}{4} \in [-\frac{1}{8}, \frac{1}{8}]$ be the displacement (in turns). We say U_N has the **no-crossing property** if $|\delta_j| < \frac{1}{8}$ for every j — equivalently, no eigenvalue sits at a cluster boundary (a gap-centre, the midpoint between adjacent fourth roots). Under no-crossing the eigenvalues partition unambiguously into four clusters, and we write $(m_1, m_i, m_{-1}, m_{-i})$ for the cluster sizes.

Conjecture 1.1 (multiplicity law). For all $N \geq 2$, U_N has the no-crossing property, and with $q = \lfloor N/4 \rfloor$,

$$(m_1, m_i, m_{-1}, m_{-i}) = \begin{cases} (q, q, q, q), & N \equiv 1, \\ (q+1, q, q, q), & N \equiv 2, \\ (q+1, q, q, q+1), & N \equiv 3, \\ (q, q-1, q, q), & N \equiv 0, \end{cases} \pmod 4.$$

This has been verified by exact computation for all $N \leq 4096$. The cluster sizes are a clean, nontrivial deformation of the Fourier multiplicities — they are not simply the Fourier multiplicities with the trivial mode removed.

The reduction. Our main theorem is that the no-crossing property — the load-bearing analytic content of Conjecture 1.1 — is equivalent to a single algebraic non-degeneracy.

Theorem 1.2 (reduction). The apex unitary U_N has the no-crossing property if and only if $\det(U_N^4 + I) \neq 0$, equivalently $\|U_N^2 - U_N^{-2}\|_{\mathrm{op}} < 2$.

Theorem 1.2 (proved in §4) compresses the spectral problem to a non-vanishing determinant, and the rest of the paper establishes the unconditional structure surrounding it: §4 records the real, positive closed form of that determinant; §5 recovers the multiplicities from the first four power traces and explains why no fixed-degree trace inequality can certify the non-vanishing; §6 locates the obstruction in a single edge eigenvalue via an explicit continuum limit; §7 proves the closed form of the field trace of $\operatorname{tr} U_N$. Section 8 collects the reduction: everything is theorem-grade except the one estimate $\det(U_N^4 + I) \neq 0$, which the numerics support with a definite margin.

Throughout, $\omega = e^{2\pi i/N}$, $\zeta_m = e^{2\pi i/m}$, and $\binom{n}{2} = n(n-1)/2$. For a normal matrix $\|\cdot\|_{\mathrm{op}}$ is the spectral radius.

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2. The apex unitary and the Fourier spectrum

We recall the minimum needed from [N1, N2]. On the mean-zero hyperplane $H_N = \{v \in \mathbb{C}^N : \sum_j v_j = 0\}$ the normalized Helmert vectors $e_k = h_k/\sqrt{k(k+1)}$ (with $h_k = (1, \dots, 1, -k, 0, \dots, 0)$, k ones) and the normalized characters $\tilde{\chi}_c = (\omega^{cj})_j/\sqrt{N}$ form two orthonormal frames. The apex unitary $U_N \in U(N-1)$ is the change of basis with entries $(U_N)_{kc} = \langle \tilde{\chi}_c, e_k \rangle = R_k C_{kc}$, where $R_k = 1/\sqrt{Nk(k+1)}$ is the row radical and $C_{kc} = \sum_{j=0}^{k-1} \omega^{cj} - k\,\omega^{ck} \in \mathbb{Z}[\omega]$ is the cyclotomic factor of [N1, Prop. 2.3]; on the diagonal $C_{kk} = \sum_{j=0}^{k-1} \omega^{kj} - k\,\omega^{k^2}$.

Let H be the full normalized Helmert matrix (row 0 the mean $\frac{1}{\sqrt{N}}(1, \dots, 1)$, rows $k = e_k$) and F the symmetric Fourier matrix $F_{cj} = \omega^{cj}/\sqrt{N}$. Then $F = F^{\mathsf{T}}$ is unitary, and:

Proposition 2.1 (N2, Prop. 3.1). $M := HF^{\mathsf{T}} = HF$ has the block form $M = 1 \oplus U_N$; in particular $\operatorname{tr}(U_N^k) = \operatorname{tr}(M^k) - 1 = \operatorname{tr}((HF)^k) - 1$ for all k, and $\operatorname{spec}(U_N) = \operatorname{spec}(M) \setminus \{1\}$.

The Fourier matrix satisfies

$$F^2 = R, \qquad F^4 = I, \qquad (Rv)_j = v_{-j} \text{ (reversal)},$$

so $\operatorname{spec}(F) \subseteq \{1, i, -1, -i\}$. Its four multiplicities depend on N mod 4 (Schur; McClellan–Parks). The eigenvalues of U_N = block of HF are the deformation of $\operatorname{spec}(F)$ studied here.

For an eigenvalue $\lambda = e^{2\pi i\theta}$ of U_N we use throughout: nearest fourth root i^c, displacement $\delta = \theta - \frac{c}{4} \in [-\frac{1}{8}, \frac{1}{8}]$, so that

$$\lambda^2 = (-1)^c e^{4\pi i\delta}, \qquad \lambda^4 = e^{8\pi i\delta}, \qquad \lambda^2 - \lambda^{-2} = 2i(-1)^c \sin(4\pi\delta), \qquad \lambda^2 + \lambda^{-2} = 2(-1)^c \cos(4\pi\delta).$$

The displacement of U_N is $\delta_{\max} = \max_j |\delta_j|$.

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3. The concentration phenomenon

Computation shows the spectrum of U_N is sharply concentrated: for every $N \leq 4096$ the eigenvalues partition into four clusters around $\{1, i, -1, -i\}$ with the multiplicities of Conjecture 1.1, and the displacements δ_j are minute for all but two eigenvalues. Precisely, the median displacement is 0 to working precision, and only **two** eigenvalues — one on the upper edge of the cluster at $+1$, one on the lower edge of the cluster at $-i$ — carry a macroscopic displacement; every other displacement is below 0.02 of a turn. The leading displacement $\delta_{\max}(N)$ grows slowly and monotonically (Table 3.1)

and, as we shall see in §6, converges to a limit $\delta_\infty \approx 0.097$, comfortably below the gap half-width $\frac{1}{8} = 0.125$.

Table 3.1. Leading displacement $\delta_{\max}(N)$ (turns) and the gap margin $\frac{1}{8} - \delta_{\max}$.

N	16	64	256	1024	4096
$\delta_{\max}$	0.0315	0.0500	0.0632	0.0727	0.0796
$\frac{1}{8} - \delta_{\max}$	0.0935	0.0750	0.0618	0.0523	0.0454

That every eigenvalue stays within the gap half-width is the no-crossing property. The remainder of the paper is organized around it.

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4. The no-crossing reduction

This section proves Theorem 1.2 and records the structure of the governing determinant. Both $U^2 \pm U^{-2}$ (we drop the subscript N) are polynomials in the unitary U, hence normal, with eigenvalues $\lambda_j^2 \pm \lambda_j^{-2}$ on the respective lines.

Theorem 4.1. The following are equivalent. 1. **(I)** U has the no-crossing property ($|\delta_j| < \frac{1}{8}$ for all j); 2. **(II)** $-1 \notin \operatorname{spec}(U^4)$, i.e. $\det(U^4 + I) \neq 0$; 3. **(III)** $(U^2 + U^{-2})^2 = 2I + U^4 + U^{-4}$ is positive definite; 4. **(IV)** $\|U^2 - U^{-2}\|_{\mathrm{op}} < 2$.

Moreover, with the Hermitian matrix $B := -i\,(U^2 - U^{-2})$ one has $B^2 = 2I - U^4 - U^{-4}$ and

$$\left\|U^2 - U^{-2}\right\| = \|B\| = 2\sin(4\pi\,\delta_{\max}).$$

Proof. On the unit circle $\lambda_j^2 - \lambda_j^{-2} = 2i\sin(4\pi\theta_j)$, $\lambda_j^2 + \lambda_j^{-2} = 2\cos(4\pi\theta_j)$, and $\lambda_j^4 = e^{8\pi i\theta_j}$.

(I)⇔(II). $\lambda_j^4 = -1 \iff 8\pi\theta_j \equiv \pi \pmod{2\pi} \iff \theta_j \in \{\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}\}$. These four points are exactly the gap-centres (midpoints between consecutive fourth roots). Hence some eigenvalue sits at a gap $\iff -1 \in \operatorname{spec}(U^4) \iff \det(U^4 + I) = 0$; negating gives (I)⇔(II).

(II)⇔(III). For $|\lambda| = 1$, $\lambda^4 = -1 \iff \lambda^2 \in \{i, -i\} \iff \lambda^2 + \lambda^{-2} = 0$ (e.g. $\lambda^2 = i \Rightarrow \lambda^{-2} = -i$). Thus $-1 \in \operatorname{spec}(U^4) \iff 0 \in \operatorname{spec}(U^2 + U^{-2})$. Since $U^2 + U^{-2}$ is Hermitian, $(U^2 + U^{-2})^2 \succeq 0$ always and is $\succ 0$ iff $U^2 + U^{-2}$ is nonsingular. The identity $(U^2 + U^{-2})^2 = 2I + U^4 + U^{-4}$ is immediate. Hence (II)⇔(III).

(III)⇔(IV). B is Hermitian because $(U^2 - U^{-2})^* = U^{-2} - U^2$. Then

$$B^2 = -(U^2 - U^{-2})^2 = 2I - U^4 - U^{-4}, \qquad 4I - B^2 = 2I + U^4 + U^{-4} = (U^2 + U^{-2})^2.$$

So $(U^2 + U^{-2})^2 \succ 0 \iff B^2 \prec 4I \iff \|B\| < 2$. Finally $\|B\| = \max_j 2|\sin 4\pi\theta_j|$; writing $\theta_j = \frac{c_j}{4} + \delta_j$ gives $|\sin 4\pi\theta_j| = \sin 4\pi|\delta_j|$ (monotone on $[0, \frac{1}{8}]$), so $\|B\| = 2\sin 4\pi\delta_{\max}$. □

The quantitative margin to crossing is therefore $2 - \|U^2 - U^{-2}\| = 2(1 - \sin 4\pi\delta_{\max})$. We next record that the governing determinant is real and positive, with an explicit value.

Proposition 4.2. $\det(U_N^4 + I)$ is a non-negative real number, and

$$\det(U_N^4 + I) = (-1)^{\binom{N-1}{2}} \det(U_N^2 + U_N^{-2}) = 2^{N-1} \prod_{j=1}^{N-1} \cos(4\pi\delta_j) \;\in\; [0,\, 2^{N-1}],$$

vanishing if and only if U_N fails no-crossing.

Proof. Factor $U^4 + I = U^2(U^2 + U^{-2})$. By the determinant theorem of [N2], $\det U^2 = (\det U)^2 = (i^{-\binom{N-1}{2}})^2 = (-1)^{\binom{N-1}{2}}$, while $U^2 + U^{-2} = U^2 + (U^*)^2$ is Hermitian with real determinant; the first equality follows. For the second, each eigenvalue contributes $\lambda_j^4 + 1 = 2\cos(4\pi\delta_j)\, e^{4\pi i\delta_j}$ with $\cos(4\pi\delta_j) \geq 0$ for $|\delta_j| \leq \frac{1}{8}$; the product of the unimodular phases is ± 1 by the reality just proved, and equals $+1$ since the product of the non-negative moduli is real and the determinant is real. Non-negativity and the range are immediate, and $\det = 0 \iff$ some $\cos(4\pi\delta_j) = 0 \iff$ some $|\delta_j| = \frac{1}{8}$. $\square$

Proposition 4.2 is the structural heart of the reduction. The determinant carries no multiquadratic radical content (it is real and, after the N2 sign, equals a Hermitian determinant), and its strict positivity is *exactly* the no-crossing property. We return in §8 to the fact that this gives no algebraic shortcut: $\det(U^2 + U^{-2})$ is the determinant of a *sum*, to which the product factorization $\det U = \det H \cdot \det F$ of [N2] does not apply.

Remark 4.3 (the multiplicity values). Theorem 4.1 establishes that no-crossing — the unambiguous fourfold clustering — is the non-degeneracy $\det(U_N^4 + I) \neq 0$. The specific cluster *sizes* of Conjecture 1.1 are an additional input; §5 shows they are recovered from the first four power traces, and are stable under no-crossing.

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5. Power-trace formulations

The multiplicities and the no-crossing condition both have transparent expressions in the power traces $p_m = \operatorname{tr}(U_N^m) = \operatorname{tr}((HF)^m) - 1$.

Proposition 5.1 (recovery of multiplicities). Under no-crossing, for $r = 0, 1, 2, 3$ (clusters $1, i, -1, -i$),

$$m_r = \Big\lfloor \tfrac{1}{4} \operatorname{Re} \sum_{m=0}^{3} i^{-rm}\, p_m \Big\rceil \quad \text{(nearest integer)}, \qquad p_0 = N - 1.$$

Proof (sketch). The quantity $a_r = \frac{1}{4}\sum_{m=0}^{3} i^{-rm} p_m = \sum_j w_r(\lambda_j)$ with $w_r(\lambda) = \frac{1}{4}\sum_{m=0}^{3}(i^{-r}\lambda)^m$. For λ in cluster r ($i^{-r}\lambda = e^{2\pi i\delta}$, small δ) one has $\operatorname{Re} w_r = 1 - 7\pi^2\delta^2 + O(\delta^4)$; for a neighbouring cluster $\operatorname{Re} w_r = O(\delta)$, and for the opposite cluster $\operatorname{Re} w_r = O(\delta^2)$. Summing, $\operatorname{Re} a_r = m_r + \varepsilon_r$ with $|\varepsilon_r| < \frac{1}{2}$ provided the displacements are small enough; the bound $|\varepsilon_r| < \frac{1}{2}$ holds throughout the computed range ($|\varepsilon_r| \leq 0.17$ for $N \leq 3000$), so the rounding returns m_r. $\square$

Proposition 5.1 reformulates Conjecture 1.1 as a rounding statement about $\operatorname{tr}((HF)^m)$, $m \leq 3$ — the leading characteristic-polynomial data. Note the *imaginary* part $\frac{1}{4}\operatorname{Im}\sum i^{-rm} p_m$ is the net displacement chirality and grows, but does not enter the rounding.

The no-crossing condition itself sits one level up, in the even moments of B. From $B^2 = 2I - U^4 - U^{-4}$,

$$\operatorname{tr}(B^2) = 2\big[(N-1) - \operatorname{Re}\operatorname{tr}(U^4)\big] = 2\,\mathcal{E}_N, \qquad \mathcal{E}_N := N - \operatorname{Re}\operatorname{tr}((HF)^4),$$

the *total displacement energy* $\mathcal{E}_N = \sum_j (1 - \cos 8\pi\delta_j)$; and more generally $\operatorname{tr}(B^{2p})$ is a $\mathbb{Z}$-combination of $\operatorname{tr}((HF)^{4k})$, $|k| \leq p$. Since $B^2 \succeq 0$,

$$\text{no-crossing} \iff \|B\| = \lim_{p\to\infty} \operatorname{tr}(B^{2p})^{1/2p} < 2.$$

Remark 5.2 (no finite-degree certificate). The lowest moment gives the sufficient condition $\mathrm{tr}(B^2) < 4$, i.e. $\mathcal{E}_N < 2$. This *fails*: the energy is shared by the two stragglers (near $+1$ and near $-i$), and $\mathcal{E}_N \to\approx 3.3 > 2$. A degree-$2p$ test $\mathrm{tr}(B^{2p}) < 4^p$ is sufficient but, because of the same two-straggler factor, certifies $\|B\| < 2$ only as $p \to \infty$ (the spectral radius $\lim_p \mathrm{tr}(B^{2p})^{1/2p} = 2\sin 4\pi\delta_\infty \approx 1.88 < 2$). Thus no single fixed-degree power-trace inequality can establish the no-crossing; the obstruction is genuinely the leading eigenvalue, not an aggregate.

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6. The edge operator

Section 5 shows the difficulty is a single edge mode. Here we identify the operator that mode belongs to, by rescaling the matrix entries. The leading straggler's eigenvector is concentrated at the low-frequency edge (indices of order $\sqrt{N}$), which fixes the scaling $k = \sqrt{N}\,u$, $c = \sqrt{N}\,v$. Substituting into the entry formula and letting $N \to \infty$,

$$\sqrt{N}\,(U_N)_{kc} \;\longrightarrow\; K(u,v) := \frac{e^{2\pi i uv} - 1}{2\pi i\, uv} - e^{2\pi i uv},$$

a limit confirmed numerically (relative error below 1% at $N = 10^4$). The corresponding operator on $L^2[0,\infty)$,

$$(\mathcal{U}\psi)(u) = \int_0^\infty K(u,v)\,\psi(v)\,dv,$$

factors cleanly: using $\dfrac{e^{2\pi i uv} - 1}{2\pi i uv} = \dfrac{1}{u}\displaystyle\int_0^u e^{2\pi i tv}\,dt$,

$$\boxed{\mathcal{U} = (A - I)\,\Phi}, \qquad (\Phi\psi)(u) = \int_0^\infty e^{2\pi i uv}\psi(v)\,dv, \quad (Ag)(u) = \frac{1}{u}\int_0^u g(t)\,dt,$$

the half-line Fourier transform Φ composed with the Hardy averaging operator minus the identity. The leading straggler is the eigenvalue of (the finite-N truncation of) $\mathcal{U}$ nearest the cluster edge; its displacement satisfies

$$\delta_{\max}(N) \;\longrightarrow\; \delta_\infty \approx 0.097, \qquad \delta_\infty - \delta_{\max}(N) \asymp N^{-1/4},$$

the exponent $\frac{1}{4}$ being the boundary-layer scaling of a box of size $\sim \sqrt{N}$. The mode is genuinely two-scale — a sharp peak of size $O(1)$ atop a body of width $\sqrt{N}$ — so δ_∞ is not the leading eigenvalue of a single-scale operator, and we do not have a closed form for it. What §6 provides is the localization: the entire obstruction of §4–§5 is the edge eigenvalue of the explicit operator $(A - I)\Phi$, and the open estimate is $\delta_\infty < \frac{1}{8}$, i.e. $2\sin 4\pi\delta_\infty < 2$.

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7. The trace and its field trace

The first power trace has independent arithmetic content. From the radical–cyclotomic split,

$$\mathrm{tr}\,U_N = \sum_{k=1}^{N-1} (U_N)_{kk} = \sum_{k=1}^{N-1} R_k\, C_{kk}, \qquad C_{kk} = \sum_{j=0}^{k-1} \omega^{jk} - k\,\omega^{k^2} \in \mathbb{Q}(\zeta_N).$$

Each diagonal entry carries a row radical $R_k = 1/\sqrt{Nk(k+1)}$. The behaviour of the radical content under the Galois group of $L_N = K_N \cdot \mathbb{Q}(\zeta_N)$ over $\mathbb{Q}(\zeta_N)$ is determined by sign characters, and is fully explicit.

Theorem 7.1. Let $\Gamma = \mathrm{Gal}(L_N/\mathbb{Q}(\zeta_N))$, of order 2^s with $s = \pi(N) - t(N)$ (notation of [N1, §2.4]), and let

$$S_N = \{\, k \in \{1, \dots, N-1\} : \sqrt{\mathrm{sf}(Nk(k+1))} \in \mathbb{Q}(\zeta_N) \,\}.$$

Then $\mathrm{Tr}_{L_N/\mathbb{Q}(\zeta_N)}\left((U_N)_{kk}\right) = 2^s (U_N)_{kk}$ for $k \in S_N$ and 0 otherwise; hence

$$\mathrm{Tr}_{L_N/\mathbb{Q}(\zeta_N)}\left(\mathrm{tr}\, U_N\right) = 2^s \sum_{k \in S_N} (U_N)_{kk}.$$

In particular the genuinely non-cyclotomic part $\sum_{k \notin S_N} (U_N)_{kk}$ has vanishing field trace; the field trace of $\mathrm{tr}\, U_N$ vanishes iff $\sum_{k \in S_N} (U_N)_{kk} = 0$.

Proof. Each $\tau \in \Gamma$ fixes $\mathbb{Q}(\zeta_N)$, hence fixes C_{kk}, and sends $R_k \mapsto \varepsilon_k(\tau) R_k$ for a sign character $\varepsilon_k \colon \Gamma \to \{\pm 1\}$ (ε_k trivial $\iff$ $R_k \in \mathbb{Q}(\zeta_N)$ $\iff$ $k \in S_N$). By orthogonality of characters $\sum_{\tau \in \Gamma} \varepsilon_k(\tau) = 2^s \, [\![k \in S_N]\!]$, so $\mathrm{Tr}_{L_N/\mathbb{Q}(\zeta_N)}(R_k C_{kk}) = C_{kk} \sum_\tau \tau(R_k) = 2^s R_k C_{kk} \, [\![k \in S_N]\!]$. Sum over k. $\square$

Theorem 7.1 corrects the naive expectation that the field trace vanishes for all N: it does so precisely when $S_N = \emptyset$ — the case at $N = 4$, where every row radical is non-cyclotomic. The first $N \geq 3$ with $S_N \neq \emptyset$ is $N = 5$, where $S_5 = \{4\}$, $(U_5)_{44} = -\frac{1}{2}\zeta_5$, and the field trace equals $-2\zeta_5 \neq 0$.

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8. The structural reduction

We collect the result. Around the spectral concentration of U_N the following are established unconditionally:

- **(Theorem 4.1)** the no-crossing property $\iff$ $\det(U_N^4 + I) \neq 0$ $\iff$ $(U_N^2 + U_N^{-2})^2 \succ 0$ $\iff$ $\|U_N^2 - U_N^{-2}\| < 2$, with $\|U_N^2 - U_N^{-2}\| = 2 \sin 4\pi\delta_{\max}$;
- **(Proposition 4.2)** $\det(U_N^4 + I) = (-1)^{\binom{N-1}{2}} \det(U_N^2 + U_N^{-2}) = 2^{N-1} \prod_j \cos 4\pi\delta_j$, real and in $[0, 2^{N-1}]$;
- **(Proposition 5.1)** recovery of the cluster multiplicities from $\mathrm{tr}(U_N^m)$, $m \leq 3$;
- **(Remark 5.2)** the no-crossing is the spectral-radius condition $\lim_p \mathrm{tr}(B^{2p})^{1/2p} < 2$, with no fixed-degree certificate;
- **(Theorem 7.1)** the closed form of the field trace of $\mathrm{tr}\, U_N$;
- **(§6)** the explicit edge operator $\mathcal{U} = (A - I)\Phi$ whose leading eigenvalue is the displacement $\delta_{\max}$.

What remains is a single estimate, in five equivalent forms:

$$\delta_{\max} < \tfrac{1}{8} \iff \det(U_N^4 + I) \neq 0 \iff \|U_N^2 - U_N^{-2}\| < 2 \iff (U_N^2 + U_N^{-2})^2 \succ 0 \iff \prod_j \cos 4\pi\delta_j > 0.$$

Granting this estimate, U_N has the no-crossing property and the spectrum partitions into the four clusters; the multiplicity values of Conjecture 1.1 then follow from Proposition 5.1. The estimate is supported numerically for all $N \leq 4096$ and, through the edge operator of §6, is localized to a single mode whose limiting displacement $\delta_\infty \approx 0.097$ lies below the threshold $\frac{1}{8} = 0.125$ with

margin ≈ 0.028. It is not reducible to a finite algebraic or trace identity: Proposition 4.2 shows the governing determinant is a determinant of a *sum*, and Remark 5.2 shows the aggregate energy provably exceeds the threshold. The concentration theorem is thus reduced to the edge estimate $\delta_\infty < \frac{1}{8}$, and no further.

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